MATT Improvement Senior Design Project

Meeting Minutes — October 15, 2026

Meeting Duration: 27 minutes, 54 seconds

Meeting Purpose: Review the existing Mission Assessment Tracking Tool (MAT), identify required functionality and current limitations, discuss user needs, and establish next steps for the senior design project.

Attendees

- Samuel Bricking — Project advisor
- Megan Hopperton — Project stakeholder and consultant
- Doug Disbennett — Former deputy assessments lead and experienced MAT administrator
- Gloria Yeomans — Internal assessment team member
- Cooper Orlowski
- Sam Anderson
- Luke Link
- Srithan Mopuru
- Noah Nguyen
- Xuhui Zou

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1. Project Background

The Mission Assessment Tracking Tool (MAT) is used during cyber exercises to track and evaluate the performance of Blue Teams as they complete cybersecurity tasks.

The system organizes evaluation criteria into a hierarchy of core tasks and more detailed measurements. It also provides a visual assessment wheel and reporting functionality to help users understand team performance.

The project aims to improve the existing system's usability, flexibility, and functionality while preserving the capabilities that are important to exercise staff.

During this meeting, Doug Disbennett demonstrated the existing system from an administrative perspective. Gloria Yeomans discussed the assessment wheel and reporting features, while Samuel Bricking outlined additional technical improvements and next steps.

2. Required Functionality to Preserve

Doug identified several existing capabilities that must remain available in the improved application.

Team Management

- Users must be able to create and manage teams within an exercise.
- The system must support multiple Blue Teams, typically Alpha, Bravo, Charlie, and sometimes Delta.
- Assessment information must remain associated with the correct team so their performance can be distinguished.
- The system should accommodate different task lists for different teams when needed.

User Management

- Administrators must be able to create users, provision and de-provision accounts, and manage user permissions.
- The system currently supports different user roles, including a White Cell Lead role with broad permissions.
- The existing user list has a display/navigation problem: it reports fewer users than actually exist and does not allow users to navigate the complete list reliably.
- Administrators often have to know a person's name and search for them to change permissions, add them, or remove them.

Improvement opportunity: Make the user list easier to navigate and make user administration more reliable.

Assessment Wheel and Task Hierarchy

- The application must support the hierarchical structure of core tasks and their associated measurements.
- Users should be able to view team performance at a high level and drill down into specific tasks to understand why a task has not been completed.
- The wheel is familiar to existing users and is considered useful, although the team has flexibility to redesign its appearance or presentation.
- Any alternative visualization must preserve the ability to explore assessment information at increasingly detailed levels.

3. Master Event Task List and Master Scenario Event List

Doug explained two important data structures used to configure exercises.

Master Event Task List (METL)

The Master Event Task List defines the tasks that a team is expected to accomplish during an exercise. These tasks represent cybersecurity and incident response activities and provide the foundation for evaluating team performance.

Master Scenario Event List (MSEL)

The Master Scenario Event List contains the events that occur during an exercise. Each event includes an event identifier, title, description of expected actions, and associated measurement codes.

The MSEL is linked to the METL so assessors can connect specific exercise events to the tasks being evaluated.

Importing Assessment Data

The current system uses an Excel file to import the event information and associated measurement codes. The imported data populates the assessment wheel and its detailed task structure.

Current limitations include:

- Measurement codes must be entered as plain text.
- Spaces or formatting issues can cause an import to fail.
- The system is restrictive about how data must be formatted before importing.
- Event identifiers currently need greater flexibility.

Requested improvements:

1. Support alphanumeric characters in event IDs rather than restricting them to numeric formats.
2. Make importing assessment data more flexible and tolerant of formatting inconsistencies.
3. Reduce the likelihood that a minor formatting issue will cause an entire import to fail.
4. Allow different task lists to be assigned to individual teams when required.
5. Preserve the relationship between imported events, measurement codes, core tasks, and the assessment wheel.

4. Assessment Visualization and Reporting

Gloria demonstrated the assessment interface from her perspective as an internal assessment team member.

She identified the team assessment wheel and downloadable team reports as useful existing features.

Assessment Wheel

The wheel provides a high-level overview of how teams are performing across their core tasks. Users should be able to drill down into specific measurements to investigate incomplete or unsuccessful tasks.

The existing wheel is familiar to users, so its usefulness should be considered before replacing it with a different visualization.

Team Reports

The application includes a feature to download reports for all teams. Gloria stated that these reports have been useful in her experience and did not recommend prioritizing changes to them without evidence that users need improvements.

However, the download function did not work during the meeting, so its current reliability needs further investigation.

Project consideration: Preserve the existing reporting capabilities unless interviews or testing identify specific problems that warrant changes.

5. Feedback From Actual Users

A major discussion point was the need to gather feedback from people who directly use MAT during exercises.

Doug and Gloria primarily interact with the administrative and internal assessment processes. Gloria emphasized that their experiences may differ from those of the exercise personnel who use the system during live activities.

The team was encouraged to interview actual users to better understand:

- Which parts of MAT are difficult or time-consuming to use.
- How users interact with the assessment wheel during an exercise.
- Which features are most important during real-time assessments.
- What problems users encounter when recording or reviewing information.
- Whether the existing reports meet their needs.
- How the system could better support both experienced and new users.

Primary User Contact

Megan identified Nate McAvoy as the initial point of contact. He has supported exercises for years and has agreed to provide feedback, subject to his availability.

The team may also consult newer users to capture perspectives that differ from those of experienced personnel.

6. Application Access and Testing

At the time of the meeting, the senior design team had not logged into or directly used the MAT application. Their understanding was based on descriptions and screenshots.

The team needs access to explore the existing system and determine which functionality should be preserved or improved.

Doug suggested that a separate project exercise could be created and assigned to the team so they could explore the system without interfering with upcoming live exercises.

Samuel and Doug agreed to investigate the possibility of creating an appropriate account or exercise for the team.

Access requirements:

- Provide the team with a suitable account, potentially with read-only permissions.
- Establish a safe environment for exploring the application's functionality.
- Avoid interfering with data or exercises used for operational purposes.
- Allow the team to examine the assessment workflow, user management, imports, and reports.

Access was not confirmed as completed during the meeting.

7. Technical Direction and Additional Features

Samuel reiterated that the team has flexibility in selecting its technology stack, provided the solution meets the project's functional requirements.

A front-end application connected to a database was discussed as a possible general architecture. However, the current technical implementation has not yet been fully examined by the team.

Database Backup and Data Export

Samuel identified database backup and export functionality as an additional improvement opportunity.

The current process requires running a back-end script, which does not always work reliably.

Potential improvements include:

- A simpler way to back up the database.
- A reliable method for exporting assessment data.
- Reduced reliance on manual back-end scripts.
- Improved confidence in data recovery and portability.

This was raised as a proposed feature, rather than a finalized requirement.

8. Decisions and Agreements

1. **Preserve essential functionality.** Team management, user administration, assessment tracking, task hierarchy, and useful reporting must be considered when designing the replacement or improved application.
2. **Improve flexibility.** Event identifiers and data imports should be less restrictive.
3. **Retain detailed assessment navigation.** Users must be able to move from high-level team performance to individual task measurements.
4. **Consult actual users.** The team should gather direct feedback before finalizing its design.
5. **Arrange application access.** Samuel and Doug will explore creating a suitable account or project exercise.
6. **Allow flexibility in technology selection.** The team can choose a stack based on its familiarity and the project's needs.
7. **Investigate data backup and export.** These are additional opportunities to improve reliability and administration.
8. **Schedule another project meeting.** Samuel planned to arrange a follow-up meeting, aiming for the same time the following week.

9. Action Items

Action Item	Responsible Party	Status
Investigate creating an appropriate MAT account or isolated project exercise	Samuel Bricking and Doug Disbennett	Pending
Arrange a meeting with Nate McAvoy	Samuel Bricking	Pending
Explore MAT directly once access is available	Senior design team	Pending
Interview experienced users about real-world workflows and pain points	Senior design team	Pending
Consider gathering feedback from newer users	Project stakeholders and senior design team	Potential follow-up

Document the current task hierarchy, assessment wheel, and import workflow	Senior design team	Pending
Investigate user-list navigation and account-management problems	Senior design team	Pending
Evaluate import validation and support for alphanumeric event IDs	Senior design team	Pending
Verify the reliability and usefulness of team report downloads	Senior design team	Pending
Evaluate database backup and export functionality	Senior design team	Pending
Select an appropriate technology stack after reviewing the existing system	Senior design team	Pending
Upload these meeting minutes to the team's GitHub repository	Cooper Orlowski	Planned

10. Open Questions for Follow-Up

The following questions remain unresolved and should be addressed as the project progresses:

- What is the existing application's architecture, programming language, and database technology?
- Will the team modify the existing application or build a replacement?
- What are the most significant usability problems experienced by assessors during live exercises?
- What permissions and access restrictions will the development team need to follow?
- How should the application validate imported files and communicate import errors?
- Are alphanumeric event identifiers the only required change to the import format, or are other structural changes needed?
- How should the system handle different task lists and measurement codes across multiple teams?
- Is the current reporting functionality reliable, and what do users need from reports?
- What backup and export formats are required?
- What are the project's final deliverables and acceptance criteria?

11. Next Steps

The immediate priority is to obtain access to MAT and arrange a conversation with Nate McAvoy. Direct system exploration and user feedback will allow the team to validate the requirements identified during the meeting.

Once access is available, the team should document the existing workflows, test the import and reporting functions, and prioritize improvements based on user needs and technical feasibility.

Key Takeaway: The team now has a clearer picture of MAT's administrative functions, assessment structure, import limitations, and reporting features. However, direct access and feedback from actual exercise users are still needed before the team can confidently finalize its scope and technical design.